

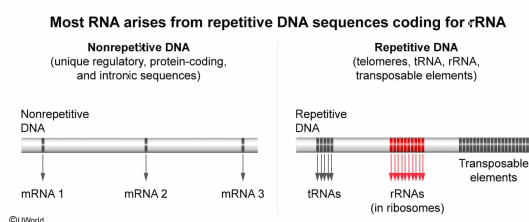
One reason that mRNA is only a small fraction of a eukaryotic cell's total RNA is presumably partly because:

- ☐ A. ribosomal RNA, as an essential component of ribosomes, is amplified after transcription. [46%]
- ✓ ☒ B. ribosomal RNA, as a transcriptional product essential for translation, is coded by repetitive DNA sequences. [48%]
- ☐ C. circular DNA is only a small proportion of a cell's total DNA. [2%]
- ☐ D. circular DNA contains telomeres that add to the cell's content of non-translated DNA. [2%]

Correct

48% answered correctly

Explanation:



The DNA in **eukaryotic genomes** can be subdivided into various classes based on characteristics such as **nucleotide** composition, sequence structure, and coding potential (ie, whether a region of DNA contains the information for producing a product). Only a small fraction of eukaryotic DNA codes for proteins (ie, **DNA → mRNA → protein**), and the percentage of genomic DNA giving rise to **different RNAs** (ie, mRNA, rRNA, tRNA) is approximately 1.5%.

Although only a small fraction of the genome codes for proteins, the expression of these protein-coding genes can be greatly augmented through multiple **rounds of translation** using the same mRNA transcript. In contrast, the **genes coding for RNA** products used in translation (ie, tRNA and rRNA) can only be **augmented through increased transcription**. Accordingly, the vast majority of a eukaryotic cell's RNA is either rRNA (~80%) or tRNA (~15%), with the remainder being mRNA.

The question asks for a reason why mRNA is only a small fraction of a eukaryotic cell's total RNA. **Most** of the **eukaryotic genome**, including that coding for the **rRNA** present in ribosomes, consists of **repetitive DNA sequences**. In the case of rRNA, these repetitive sequences allow cells to produce the millions of ribosomes necessary for translation.

(Choice A) Ribosomal RNA is an essential component of ribosomes; however, if RNA is amplified, it occurs *via* transcription (*not* after transcription).

(Choice C) Circular DNA (eg, mitochondrial DNA) is a small portion of a cell's DNA. However, because mRNA arises from both nuclear and mitochondrial DNA, the fact that most DNA is not circular does not explain why mRNA is only a small proportion of a cell's total RNA.

(Choice D) Circular DNA does not have ends; therefore, it does not contain **telomeres**.

Educational objective:

The majority of a eukaryotic cell's DNA consists of repetitive sequences. Some of these sequences are known to give rise to gene products (RNA or proteins) whereas most repetitive sequences correspond to transposable elements.

Time spent:0 sec**Subject:**

Biology

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Molecular Biology